

- 5 The random variable X has probability generating function $G_X(t)$ given by

$$G_X(t) = k(1 + 3t + 4t^2),$$

where k is a constant.

- (a) Show that $E(X) = \frac{11}{8}$. [3]

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The random variable Y has probability generating function $G_Y(t)$ given by

$$G_Y(t) = \frac{1}{3}t^2(1 + 2t).$$

The random variables X and Y are independent and $Z = X + Y$.

- (b) Find the probability generating function of Z , expressing your answer as a polynomial in t . [2]

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- (c) Use your answer to part (b) to find the value of $\text{Var}(Z)$. [3]

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- (d) Write down the most probable value of Z . [1]

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